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Train to Wayside Communication-VF.CONCEPT

SFRS for Zefiro-Europe

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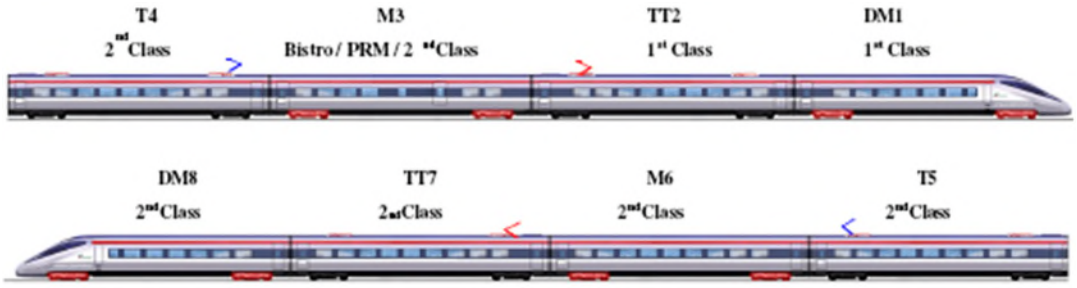
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1 Introduction

This document is a function specification for the control of the Telediagnostica System (TWCS) for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
TWCS	Train to Wayside Communication System
MCG	Mobile Communication Gateway
TU	Telediagnostica Unit
RSMS	Rolling Stock Management System
GPRS	General Packet Radio Service
GSM-R	Global System for Mobile Communication - Railway
WiFi	Wireless Fidelity
ODBS	Onboard DataBase System
OTI	Offboard Tool Interface
ECN	Ethernet Consist Network
DDS	Diagnostic Data Set
FSM	Finite State Machine
WLAN	Wireless Local Area Network
CSS	Common Software Structures (Real Time O.S.)
RSMS	Rolling Stock and Management System

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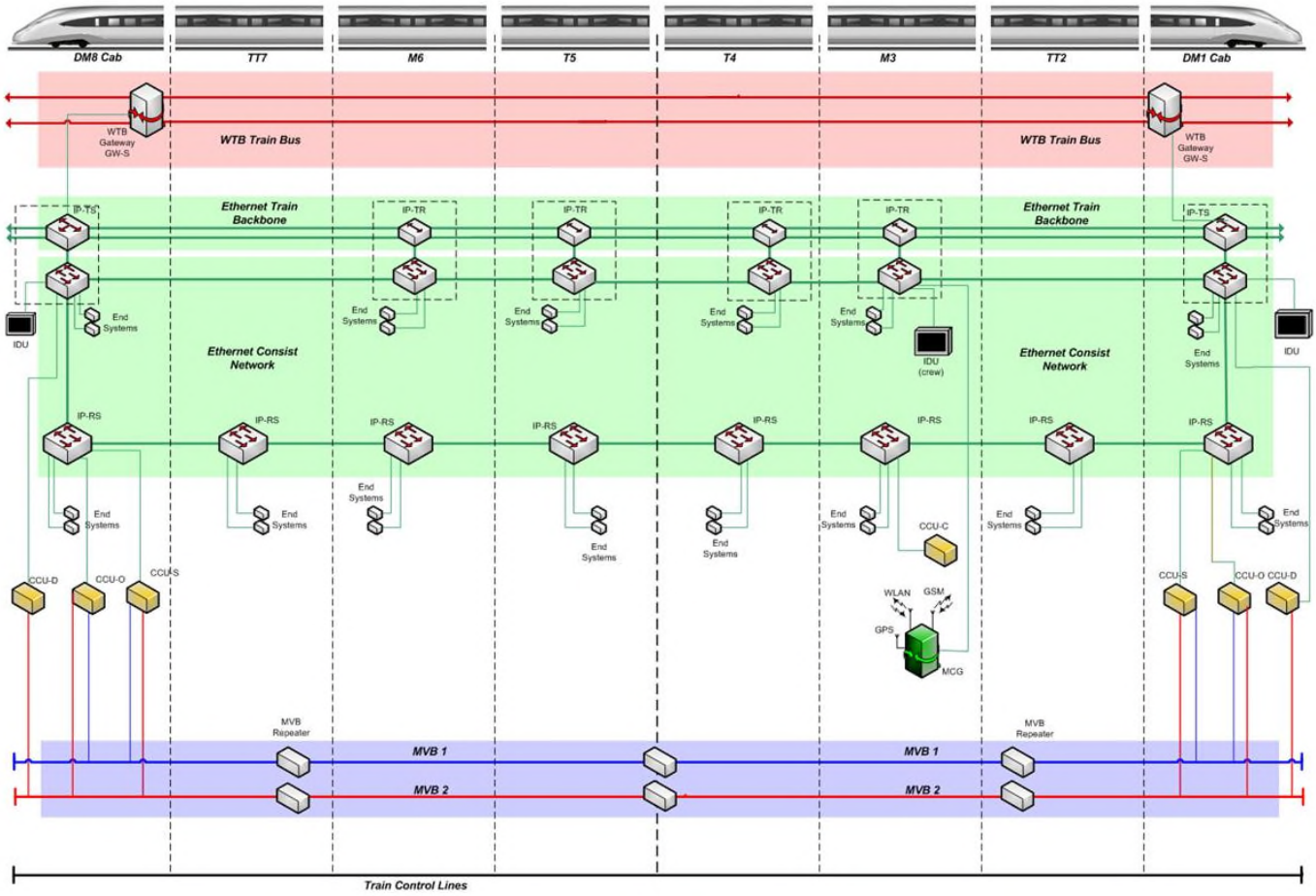
2 System Overview

Telediagnostica system can be schematically subdivided in the following subsystem:
On-board System consists of two devices for each train: MCG that manages the communication with wayside system and TU which, interfaced with the TCMS, acquires, processes and makes the diagnostics available to the wayside system;
Wayside System consists of:
Wayside Server: cluster of servers that filter, process and aggregate the data received by the on-board system. Basing on diagnostic rules, it generates alerts for the maintenance stuff with the predisposition to be interfaced to Trenitalia's RSMS.
Maintenance Tool: it is a set of web and desktop applications that allow processing the diagnostic data in off-line condition, also with the help of statistical analysis tools.
Communication between on-board and wayside systems is bi-directional and it uses two different channels:
a. Wi-Fi in depot or in stations provided with access point of the customer intranet.
b. GPRS over GSM-R where WiFi is not available.
Telediagnostica system collects and analyses all the data generated by the train diagnostics. Data is sent, using the wireless communication channels, to the wayside server where it's stored on a central database. The wayside server, through the application of diagnostic and statistical algorithms to the data received from the on-board system, identifies the failures and/or the malfunctions and, where possible, the causes.
Telediagnostica system is able to support the automatic generation of work orders for the maintenance department basing on the application of diagnostic algorithms. Work orders contain all the information and indication useful to allow the maintenance staff to identify the defected component/subsystem in an efficient way. The work orders may be transferred automatically by the telediagnostica system to Customer's RSMS system.

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2.1 System Configuration



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3 Functional requirements

3.1 Function Self Diagnosis

3.1.1 Function Design

MCG and CCUC devices perform autotest and publish status information on ECN.

3.1.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.164
Description	During boot-up, the device will perform diagnosis activities of its main components of the on board system (boards, RAM, etc.).
Safety level	

ID	2F_12.Zefiro-Europe.TRS.165
Description	Moreover, all active system processes will be cyclically checked on the basis of configuration, after minor failures and, if necessary, restarted. In case of serious non-recoverable damage the complete system will be reset or a controlled shutdown will be performed.
Safety level	

ID	2F_12.Zefiro-Europe.TRS.166
Description	The device does not require manual activities to restore non-blocking hardware or software failures of its component.
Safety level	

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.423	MCG shall publish an IPTCom PD including a life signal.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.419	MCG life signal shall be supervised by TCMS. If it is not incremented for 2 seconds, an event shall be set in the diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_12.Zefiro-Europe.CONCEPT.418	MCG shall publish an IPTCom PD that includes the communication engines status (GSM-R and WiFi).	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.420	Communication engines status shall be supervised by TCMS. On communication engine error an event shall be set in the diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_12.Zefiro-Europe.CONCEPT.593	CCUC shall publish an IPTCom PD including a life signal.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		
2F_12.Zefiro-Europe.CONCEPT.594	CCUC shall publish an IPTCom PD including the result of the autotest.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		
2F_12.Zefiro-Europe.CONCEPT.595	CCUC shall supervise running processes and restart them on failure.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		
2F_12.Zefiro-Europe.CONCEPT.411	When ODBS reaches a critical filling level, a signal shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.2 Function Event Data and Condition Data upload

3.2.1 Function Design

TCMS diagnostics is stored in two redundant TDS2 ODBS. MCG extracts event data from ODBS every 5 minutes and upload it to wayside servers as files using BT internal standard OTI format. After the MCG upload, events are still available in the train ODBS to be retrieved, if necessary, by the train crew using TDS Uploader tool.
ODBS manages its non volatile memory as a ring buffer: oldest DDS are overwritten when no more free space is available. MCG deletes OTI files after they are successfully uploaded to wayside server.
To avoid overwriting of diagnostic data not yet uploaded to wayside, CCUC performs a backup of ODBS events.
CCUC extracts events and condition data from ODBS cyclically and archives them as files. The files are kept on device for 10 days in order to be accessible by train crew via maintenance laptop (FTP transfer) or USB stick. Diagnostic data on CCUC is not accessible from HMIs for the on-line visualization.

3.2.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.85
Description	The on-board system shall read the content of diagnostic database (ODBS) to upload it to the wayside system.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.56
Description	The on-board unit shall upload diagnostic data (and eventually states and alarms) using GPRS and WiFi communication interfaces.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.102
Description	The on-board system shall archive TCMS diagnostic data in order to be uploaded manually using an USB stick.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.110
Description	The on-board unit shall remove from its permanent memory diagnostic data files already uploaded to the ground server.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_12.Zefiro-Europe.CONCEPT.416	MCG device shall export event data and condition data from the odbs as oti files in order to upload them on wayside servers.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.410	MCG device shall upload files on wayside servers using GPRS communication interface.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.604	MCG device shall upload files on wayside servers using WiFi communication interface when available in depot or train stations.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		
2F_12.Zefiro-Europe.CONCEPT.537	MCG device shall remove oti files when successfully uploaded on wayside servers.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.397	MCG device shall perform upload event data requests cyclically each 5 minutes.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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2F_12.Zefiro-Europe.CONCEPT.534	MCG device shall perform upload condition data requests cyclically each 60 minutes.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.389	CCUC device shall export event data and condition data from odbs as files.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.605	CCUC shall export event data ciclically each 5 minutes.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		
2F_12.Zefiro-Europe.CONCEPT.606	CCUC shall export condition data ciclically each 60 minutes.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		
2F_12.Zefiro-Europe.CONCEPT.271	CCUC shall keep event data and condition data files on filesystem for 10 days in order to avoid loss of data in case of MCG-wayside communication permanent failure or in order to allow train crew to upload it using USB stick or a maintenance laptop.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.417	Telediagnostica on-board system shall communicate with ODBS using TDS Tool protocol (3EGM081220D9008).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_12.Zefiro-Europe.CONCEPT.396	When a new event data is generated or an existing event becomes passive (event closure) ODBS shall increment the change count signal.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.3 Function Remote monitoring in real time

3.3.1 Function Design

Telediagnostica on-board system is able to start a data stream of configurable sets of process variables on request from wayside servers.
Main goal of this feature is to visualize on the website the state of the driver desk (virtualizing) in order to check the status of controls, indicators, displays in the driving cab.
This function also allows authorized maintainers to get real time access to a graphical representation of the vehicle status and navigate through HMI pages.

3.3.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.51
Description	The maintainers shall get remote access to a representation of the active train driver desk status in real time.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.58
Description	The maintainers shall get remote read access to a representation of the active cab HMI in real time.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.59
Description	The maintainers shall get read access to all available process data on the TCMS and propulsion system using DCUTerm tool (Only by wired connection).
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.67
Description	The maintainers shall get read access to parameters in the TCMS and propulsion system using the DCUTerm tool (Only by wired connection).
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.60
Description	The refresh time of signals transferred to wayside for remote driver desk/HMI session shall be scaled on the available bandwith and priority management.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.69
Description	The maintainers shall get read access to current diagnostic status of the train.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.291	Telediagnostica on-board system shall activate, on request, a process data streaming in order to allow the telediagnostica website to represent the active driver desk status in real time.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.153	Telediagnostica on-board system shall activate, on request, a process data streaming in order to allow the telediagnostica website to represent the active cab HMI in real time.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.551	CCUC device shall stop the process data streaming in case of MCG-wayside communication interruption.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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2F_12.Zefiro-Europe.CONCEPT.552	CCUC shall send to wayside server status messages to allow telediagnostica website to represent the connection status.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.553	CCUC shall send messages to wayside servers to allow telediagnostica website to represent the GPS position of the vehicles.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.433	CCUC device shall be included in sink to the multicast group grpCCUOC.aCar.lCst.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_12.Zefiro-Europe.CONCEPT.145	TCMS shall publish on ECN all signals needed to characterize the driver desk status (including HMI pages) to a dedicated IPTuri (grpCCUOC.aCar.lCst) with a refresh time of 1 second.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.4 Function Process data logging

3.4.1 Function Design

CCUC device provides the possibility to define process data recordings and to upload it using MCG device. Results files are processed and interpreted by wayside servers and shown in a user friendly way on the telediagnostica website.
Using the website, it is also possible to select the vehicle or the fleet of vehicles on which apply data loggings.
Following the list of configurable parameters:

- The list of signals available on ECN or MVB busses,
- The sampling time,
- The logging duration,
- The logical condition based on signals that starts the data logging,
- The logical condition based on signals that stops the data logging,
- The time validity (periodicity).

3.4.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.112
Description	The on-board unit shall be able to log sets of signals acquired from the vehicle busses.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.123
Description	Data logging results shall be uploaded on wayside servers.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.125
Description	It shall be possible to configure data logging sample time
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.126
Description	It shall be possible to configure data logging duration
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.127
Description	It shall be possible to configure data logging periodicity
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.128
Description	It shall be possible to configure data logging application (fleets of vehicles or individual vehicles)
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.129
Description	It shall be possible to configure data logging triggering start conditions
Safety level	N/A

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Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.542	CCUC device shall be remotely configurable in order to log sets of signals acquired from vehicle busses.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.543	CCUC device shall upload process data loggings on wayside servers.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.544	The sample time of the process data loggings shall be configurable.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.545	The duration of the process data loggings shall be configurable.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.546	The periodicity of the process data loggings shall be configurable.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.547	The application (fleets of vehicles or individual vehicles) of the process data loggings shall be configurable.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.557	It shall be possible to define arithmetic conditions based on process variables acquired from train busses in order to trigger process data loggings start and stop.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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3.5 Function Operating Counters

3.5.1 Function Design

This functionality provides the possibility to remotely configure CCUC from telediagnostica website in order to define new condition data and to define algorithms to be executed in order to increment it.

For new Condition Data it is possible to configure:
- logical conditions to increase new counters basing on process variables
- increasing expressions

New condition data are exported as standard oti files and uploaded on wayside servers using MCG device.
It is possible to dispatch new condition data on a vehicle, a group of vehicles or the whole fleet.

3.5.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.135
Description	The on-board system shall allow to remotely configure new condition data.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.554	CCUC device shall allow to remotely configure new condition data using telediagnostica website.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.556	It shall be possible to configure arithmetic conditions based on process variables available on train busses in order to trigger the increase of the counters.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.558	It shall be possible to configure the increasing amount using arithmetic expressions based on process variables available on train busses.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.609	It shall be possible to select the fleet of vehicles (one single vehicle, a fleet of vehicles or the whole fleet) on wich apply new operating counters.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.607	CCUC device shall upload new operating counters on wayside server as files ciclically each 60 minutes.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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3.6 Function Execution of diagnostic algorithms

3.6.1 Function Design

This functionality provides the possibility to configure CCUC device remotely in order to execute algorithms able to generate new event data.

For new Event Data it is possible to configure:
- sets of environmental data
- logical conditions to activate new events basing on process variables
- logical conditions to deactivate new events basing on process variables
- time validity of the algorithms

New event data are exported as standard oti files and uploaded on wayside servers using MCG device.

3.6.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.158
Description	The on-board unit shall be able to execute specific diagnostic algorithms in order to generate diagnostic events triggered by programmable conditions.
Safety level	

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.567	CCUC device shall be remotely configurable from telediagnostica website to execute diagnostic algorithms in order to generate new diagnostic events.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.568	It shall be possible to configure arithmetic conditions basing on process variables available on train busses in order to activate new diagnostic events.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.569	It shall be possible to configure arithmetic conditions basing on process variables available on train busses in order to deactivate new diagnostic events.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.570	It shall be possible to configure new sets of environmental data to attach to new diagnostic events.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.571	CCUC device shall upload new diagnostic events on wayside server as files ciclically each 5 minutes.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.610	It shall be possible to define the fleet on which apply the rule. (One vehicle, a fleet of vechiles or the whole fleet)	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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3.7 Function Upgrading of SW configuration

3.7.1 Function Design

TWCS functionalities are remotely configurable from telediagnostica website.
Configuration files are transferred from the wayside servers to the CCUC device using the MCG standard framework.
Configurations are versioned.
Versions are exported by MCG device using the Version Reporting System and uploaded to wayside servers in order to be shown on the telediagnostica website.

3.7.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.138
Description	It shall be possible to change the on-board system configuration.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.555	CCUC shall allow the update of the configuration remotely from telediagnostica website.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.575	CCUC shall report the version of the configuration in order to be shown on telediagnostica website.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.582	MCG shall upload on wayside servers Version Reporting System files cyclically each 4 hours.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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3.8 Function Connection to maintenance laptop

3.8.1 Function Design

This functionality allows the train crew to connect a maintenance PC to the ECN and download archives of event data, of condition data and of process data loggings. Using the maintenance laptop, train crew is also able to have direct access to process data available on train busses and start a session of real time data streaming.

3.8.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.145
Description	It shall be possible to update the configuration of the on-board system using a maintenance PC equipped with proper management software.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.146
Description	It shall be possible to access to data logging files using a maintenance laptop equipped with proper management software.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.147
Description	It shall be possible to access signals monitoring in real time using a maintenance laptop equipped with proper management software.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.538	CCUC device shall allow the update of the configuration using a maintenance laptop equipped with MTVD tool.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.539	MGC device shall allow the update of the configuration using a maintenance laptop equipped with MTVD tool.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.541	CCUC device shall allow the signal monitoring in real time to a maintenance laptop equipped with DCUTerm tool.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.548	CCUC device shall connect to DCUTerm tool in order to allow the mainteners to get read access to parameters available on train busses.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.540	CCUC device shall allow the upload of process data loggings using a maintenance laptop equipped with the ftp server of DCUTerm tool.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.580	CCUC device shall allow the upload of new operating counters (new condition data) using a maintenance laptop equipped with the ftp server of DCUTerm tool.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.581	CCUC device shall allow the upload of new event data using a maintenance laptop equipped with the ftp server of DCUTerm tool.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.533	Diagnostic data already uploaded by MCG shall still be available in odbs in order to be uploaded using a maintenance laptop equipped with TDS-Uploader tool.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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2F_12.Zefiro-Europe.CONCEPT.342	The maintainers shall get read access to all available process data on the TCMS and propulsion system on the train using the DCUTerm tool.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_12.Zefiro-Europe.CONCEPT.454	The maintainers shall get read access to parameters for maintenance activities in the TCMS and propulsion system using the DCUTerm tool	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.9 Function Downloading data using USB stick

3.9.1 Function Design

The train crew have the possibility to upload archives of event data, of condition data and of data loggings simply plugging-in a USB stick in CCUC device.

3.9.2 Requirements

ID	2F_12.Zefiro-Europe.TRS.150
Description	The vehicle diagnostic shall also be uploaded using a removable memory with USB interface, just inserting it in the on-board unit.
Safety level	N/A

ID	2F_12.Zefiro-Europe.TRS.151
Description	The signals data logging shall also be uploaded using a removable memory with USB interface, just inserting it in the on-board unit.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.559	CCUC device shall automatically upload event data exported from odbs on USB stick when plugged-in.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.566	CCUC device shall automatically upload condition data exported from odbs on USB stick when plugged-in.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.560	CCUC device shall automatically upload data loggings on USB stick when plugged-in.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.564	CCUC device shall automatically upload new operating counters (new condition data) on USB stick when plugged-in.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.565	CCUC device shall automatically upload new event data generated by execution of diagnostic algorithms on USB stick when plugged-in.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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3.10 Function CBM data upload

3.10.1 Function Design

CCUC provides the functionality to request predefined CBM elaborations to CBM relevant units (e.g. bogie, HVAC etc.).
CCUC generates output files with relative responses and upload it on wayside server using MCG in order to make it available on telediagnostica website.
CCUC is also able to generate output files with elaborations spontaneously sent by CBM units.

3.10.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_12.Zefiro-Europe.CONCEPT.572	CCUC device shall be remotely configurable in order to trigger predefined CBM elaborations to CBM relevant units.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.578	CCUC shall collect CBM elaborations responses and upload them on wayside servers.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A
2F_12.Zefiro-Europe.CONCEPT.579	CCUC device shall also collect CBM elaborations spontaneously sent from CBM units and upload them on wayside servers.	Derived Requirement	4S_09.03.01.01.-.Train To Wayside Communication Systems		N/A

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Outputs:

4.1.2 TCMS SW

Inputs:

Outputs:

4.2 Bus Interface

Telediagnostica to TCMS SW:

CCUC to TCMS SW

Nr	Functional name	Type	Linked to requirement
2F_12.Zefiro-Europe.CONCEPT.598	CCUC Status Data - ISystemLifeSign (dataset/comid numbering defined in 3ECP410162-0001_Ben TWCS_CCUC_TCMS_ICD)		2F_12.Zefiro-Europe.CONCEPT.593 2F_12.Zefiro-Europe.CONCEPT.594
2F_12.Zefiro-Europe.CONCEPT.599	CCUC Status Data - ISystemStatus (dataset/comid numbering defined in 3ECP410162-0001_Ben TWCS_CCUC_TCMS_ICD)		2F_12.Zefiro-Europe.CONCEPT.594
2F_12.Zefiro-Europe.CONCEPT.600	CCUC Status Data - IMobadStatus (dataset/comid numbering defined in 3ECP410162-0001_Ben TWCS_CCUC_TCMS_ICD)		2F_12.Zefiro-Europe.CONCEPT.594
2F_12.Zefiro-Europe.CONCEPT.601	CCUC Status Data - ICFlashStatus (dataset/comid numbering defined in 3ECP410162-0001_Ben TWCS_CCUC_TCMS_ICD)		2F_12.Zefiro-Europe.CONCEPT.594
2F_12.Zefiro-Europe.CONCEPT.602	CCUC Status Data - INVRamStatus (dataset/comid numbering defined in 3ECP410162-0001_Ben TWCS_CCUC_TCMS_ICD)		2F_12.Zefiro-Europe.CONCEPT.594

MCG to TCMS SW

Nr	Functional name	Type	Linked to requirement
2F_12.Zefiro-Europe.CONCEPT.373	MCG Status 1 - LifeSignal (dataset/comid numbering defined in 3EH-300300-0024 TWCS_MCG_TCMS_ICD)		2F_12.Zefiro-Europe.CONCEPT.423
2F_12.Zefiro-Europe.CONCEPT.453	MCG Status 1 - MCGStatEngines (dataset/comid numbering defined in 3EH-300300-0024 TWCS_MCG_TCMS_ICD)		2F_12.Zefiro-Europe.CONCEPT.418

TCMS SW to Telediagnostica:

TCMS SW to MCG

TCMS SW to CCUC

Nr	Functional name	Type	Linked to requirement
2F_12.Zefiro-Europe.CONCEPT.357	ODBS Status Output - EDChangeCount [ComId 220] (dataset/comid numbering defined in 3EGM081240D9001 TDS_ODBS_Users_Manual)		2F_12.Zefiro-Europe.CONCEPT.396
2F_12.Zefiro-Europe.CONCEPT.452	ODBS Status Output - DDBFull [ComId 220] (dataset/comid numbering defined in 3EGM081240D9001 TDS_ODBS_Users_Manual)		2F_12.Zefiro-Europe.CONCEPT.411

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